

INDIAN STATISTICAL INSTITUTE

Mid-Semester Examination Data Structures and Algorithms

December 12, 2025

Total Marks: 50

Note: You may answer any part of any question, but the maximum score you can obtain is 50. For any algorithm you design, you must write a proof of its correctness and derive the running time complexity.

1. Prove that the solution to the recurrence relation $T(n) = aT(n/b) + n^k$, where a and b are integer constants, $a \geq 1$, $b \geq 2$, and k is a positive constant, is:

$$T(n) = \begin{cases} O(n^{\log_b a}) & \text{for } a > b^k \\ O(n^k \log n) & \text{for } a = b^k \\ O(n^k) & \text{for } a < b^k \end{cases} \quad (15)$$

2. You have 10000 numbers coming one by one. The numbers are of two types. As the numbers come, you have to store them according to their type. You may also have deletion requests specifying a type. For a deletion request, you should delete the latest received number of that type that you have.

Write the pseudocode to implement this on a single array of size 10000. (15)

3. Write an $O(n)$ time algorithm for finding the median of n numbers. (15)

4. The inorder and postorder traversals of a binary tree are as follows:

(a) **Inorder:** 40, 50, 20, 70, 80, 10, 30, 90, 60, 100, 110, 120

(b) **Postorder:** 40, 50, 70, 80, 20, 90, 100, 110, 120, 60, 30, 10

Construct the binary tree. Show all the intermediate steps. (15)